

C language, structure of a program and Turbo C IDE ★ Golden

One college period · 40–45 minutes · Programming using C

What you will learn

- Write the basic structure of a C program
- Explain #include and #define with examples
- Use Turbo C shortcuts and know compiler vs editor vs debugger
- List reserved words and rules of identifiers

C was designed by Dennis Ritchie at Bell Labs (1972) to write UNIX. It is a middle-level, compiled, case-sensitive language. BIEK still expects Turbo C style in many colleges: #include <stdio.h>, #include <conio.h>, void main(), clrscr(), getch().

```
#include <stdio.h>
#include <conio.h>
#define PI 3.14
void main()
{
    clrscr();
    printf("Board of Intermediate Education Karachi\n");
    getch();
}
```

Part	Job
#include <stdio.h>	Preprocessor: copies standard I/O functions (printf, scanf)
#include <conio.h>	Turbo C console: clrscr, getch
#define PI 3.14	Manifest constant — name replaced before compile
void main()	Execution starts here (ISO C prefers int main(void))
Declarations	Variables before use in C89 / Turbo C
Statements	End with semicolon
getch()	Wait for a key so the output screen stays

Identifiers and keywords

- Letters, digits, underscore; cannot start with a digit; no spaces or special symbols; cannot be a keyword.
- Valid: std_marks, NUMBER5, Number_5. Invalid: 5number, 1st number, void, int, float.
- Keywords (reserved): auto, break, case, char, const, continue, default, do, double, else, enum, extern, float, for, goto, if, int, long, register, return, short, signed, sizeof, static, struct, switch, typedef, union, unsigned, void, volatile, while.

Escape sequence. A backslash code inside a string: \n new line, \t tab, \\ backslash, \a beep, \" quote, \0 string terminator.

HOW BIEK ASKS THIS

2025/26 shorts: structure of C; #include or #define; reserved words; valid/invalid identifiers; why C is case sensitive; IDE shortcuts.

اردو میں سمجھیں

C پروگرام #include سے شروع ہوتا ہے، main() سے چلتا ہے۔ #define نام کو قدر سے بدل دیتا ہے۔ (int, Reserved words void, for نام نہیں بن سکتے۔)

CHECK YOURSELF

1. Who developed C?

2. What does \n do?

Answers: 1. Dennis Ritchie · 2. Moves the cursor to a new line

MCQs

1. C programs are converted to machine language by a:
 - a. Editor
 - b. Assembler
 - c. Compiler
 - d. Debugger
2. This is not a valid identifier:
 - a. Number
 - b. 5number
 - c. NUMBER5
 - d. Number_5
3. Comments in C are enclosed in:
 - a. /* ... */
 - b. /* ... /*
 - c. */ ... /*
 - d. ** **
4. Names of variables and functions are called:
 - a. header files
 - b. identifiers
 - c. loops
 - d. structures

Key: 1=c, 2=b, 3=a, 4=b

Short questions (3 marks each)

1. Write the basic structure of a C program.
2. Define #include or #define with two examples.
3. What is an IDE? Write a few shortcut keys.
4. Identify valid/invalid: void, std_marks, 1st number.
5. Why is C called case-sensitive? Define three reserved words.

Model short answers

1. Preprocessor directives, global declarations, main(), local declarations, statements, closing brace.
Show a 8-line skeleton as above.
2. #include <stdio.h> brings printf. #include <math.h> brings sqrt. #define PI 3.14 and #define MAX 100 create constants.
3. IDE = editor+compiler+debugger. Turbo C: F2 save, F9 compile, Ctrl+F9 run, Alt+F5 user screen, F8 step.
4. void invalid (keyword). std_marks valid. 1st number invalid (digit start + space).
5. TOTAL and total are different. Keywords: int, return, while.

Long questions (board Section C)

1. Explain the structure of a C program with a labelled example. Discuss tokens: keywords, identifiers, constants, operators.